MIL-I-38121B(USAF)
AMENDMENT 3
15 FEB 84
Superseding
AMENDMENT 2
18 September 1980

MILITARY SPECIFICATION

INDICATOR, RATIO, ENGINE PRESSURE, EGU-9A

This amendment forms a part of Military Specification MIL-I-38121B(USAF), dated 31 December 1970, and is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

*2.1: Under "SPECIFICATIONS, Federal", delete "L-P-378" and "PPP-C-1120" and titles thereto. Add the following:

"PPP-B-601 Boxes, Wood, Cleated Plywood"

Under "SPECIFICATIONS, Military", delete "MIL-E-5272" and title thereto.

*Add the following boxed paragraph to bottom of page:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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*Under "STANDARDS, Military", add the following:

"MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-810 - Environmental Test Methods"

*3.1 Delete and substitute:

"3.1 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.4)."

PAGE 3

3.5.3, second sentence: Delete and substitute "The equipment lower test Mean-Time-Between-Failure (MTBF) shall not be less than 1200 hours."

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*3.6.1,d: Delete and substitute:

"d. High altitude and low temperature, as specified."

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*4.2,a: Delete and substitute "a. First article inspection (see 4.4)".

*4.4: Delete and substitute:

"4.4 First article inspection. The first article tests identified in Table III shall be performed as described under 4.6."

*4.4.2, line 1: Delete "qualification" and substitute "first article".

4.5: Delete and substitute the following:

"4.5 Quality conformance inspection. The quality conformance tests identified in Table III shall be performed as described under paragraph 4.6 and consist of the following:

a. Individual tests
b. Sampling plans and tests"

4.5.1: Add the following:

"d. Position error
e. Integral lighting, when applicable
f. Dielectric strength"

4.5.2.1: Delete "f. Dielectric strength".

4.5.2.2: Delete "d. Torque" and add:

"d. Extreme low temperature exposure
e. Extreme high temperature exposure
f. Salt Spray
g. Reliability"

*4.6.2,a, line 2: After "S_1 and" add "S_3".

*4.6.4, line 2: Delete "procedure XII, curve B, of MIL-E-5272." and substitute "method 514.2, equipment category b.1, procedure I (part 1 only), curve B of MIL-STD-810."

*4.6.14, line 2: Delete "procedure I of MIL-E-5272" and substitute "method 509.1, procedure I of MIL-STD-810".

*4.6.15: Delete and substitute the following:

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"4.6.15 Reliability. The indicator shall be subjected to a reliability demonstration at the following conditions of temperature, vibration, input voltage and the cycling of these conditions to be applied to the equipment during the test:

a. Temperature: -55°C to +85°C (-67°F to +185°F)

b. Temperature cycling: Temperature cycling shall be time to stabilize at low temperature followed by time to stabilize at the high temperature, plus two hours. Stabilize the equipment temperature at the lower test level temperature. Turn the equipment on and allow it to experience the combined effect of chamber heating and equipment operation until the point of maximum thermal inertia stabilizes at the higher temperature as determined in accordance with the thermal survey. The thermal survey shall have been previously conducted in accordance with paragraph 5.1.5 of MIL-STD-781. Maintain the equipment at the upper test level temperature for a period of two hours. Turn the equipment off and allow it to experience chamber cooling until the point of maximum thermal inertia stabilizes at the lower test level temperature. This cycle shall continue for the duration of the test.

c. Vibration: 2.0G ± 10% peak acceleration value at any nonresonant frequency between 20 and 60 Hz measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time. When the equipment being tested contains circuit boards or cards, vibration shall be normal to the plane of the majority of the cards. Otherwise, the direction of vibration is not critical. This vibration requirement may be met by mounting the equipment solidly, (that is, without the equipment shock vibration mounts) to a strong flat plate which is supported by vibration isolators. The desired amplitude and frequency may be obtained by a simple vibration machine with an asymmetric weight on a shaft attached to the plate and driven by a suitable motor. It is recommended that the vibration equipment be checked for proper operation each 24 hours of operation, and that the vibration transducer be "on" and monitored continuously during vibration.

d. Equipment on-off cycling: Equipment off during cooling cycle and on during heating cycle.

e. Input voltage: Nominal specified voltage +5 -2%.

f. Input voltage cycling: Does not apply.

The accept-reject criteria shall be that of Test Plan IC of MIL-STD-781. The lower test MTBF (θ_1) shall not be less than 1200 hours. The equipment shall be operated through the entire performance profile specified herein during the Reliability Demonstration Test."

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*4.6.20: Delete and substitute the following:

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"4.6.20 High altitude-low temperature. The indicator shall be installed in a temperature, altitude chamber equal to MIL-STD-810 in a manner similar to its normal mounting on an aircraft instrument panel. The high-altitude low-temperature test shall be conducted at a temperature of -55°C and with a chamber pressure equivalent to an altitude of 50,000 feet for a period of 4 hours. While under these conditions and at the end of the 4-hour period, a scale and friction error test shall be conducted. The scale errors shall not exceed the tolerances specified in tables I and II."

5: Delete 5. through 5.3 and substitute the following:

"5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified.

5.1.1 Level A.

5.1.1.1 Cleaning. Indicators shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Indicators shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the procuring activity, each pressure indicator, EGU-9/A, shall be packaged in quantity unit packs of one each in accordance with Method IC-1 of MIL-P-116. Overbox in PPP-B-636 carton large enough to allow for application of sufficient cushioning material between container and bag, of a type, density and thickness to insure shock transmission does not exceed peak values in Gs established for the item when completed packs are subjected to the rough handling drop tests of MIL-P-116.

* 5.1.2 Level C. Indicators shall be clean, dry and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from source of supply to the first receiving activity.

* 5.2 Packing. Packing shall be level A, B or C as specified.

* 5.2.1 Level A. Indicators packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, styles A and B, class overseas, unless otherwise specified by the procuring activity. Insofar as practical, exterior shipping containers shall be of uniform shape, size and of minimum tare and cube, consistent with protection required.

* 5.2.2 Level B. Indicators packaged as specified in 5.1.1 shall be packed in weather resistant class exterior shipping containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 above are applicable.

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* 5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first domestic receiving activity for immedite use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level.

* 5.3 Marking. In addition to any special marking required by the contract or order, each unit package and shipping container shall be marked in accordance with MIL-STD-129."

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*6.2: Add the following:

"e. Whether first article inspection is required.

*6.4 Delete entirely and substitute the following:

"6.4 First article. When a first article is required, the item will be inspected and tested in accordance with Section 4 of this document. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article."

Add new paragraph 6.5 as follows:

* "6.5 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item."

NOTE: The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the prevoius amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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